

Zhenhao Ni

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EDUCATION

Soochow University

B.Eng. in Robotics Engineering; GPA: 3.7/4.0

Sep. 2022 – Jul. 2026

Suzhou, China

RESEARCH INTERESTS

Humanoid Loco-Manipulation • Foundation Models for Robotics • Scalable Data Generation for Robot Learning

SELECTED RESEARCH & PUBLICATIONS

SIMPLE: Simulation-Based Policy Learning and Evaluation for Humanoid Loco-Manipulation

Co-first author • CoRL 2026 • [Project] [Code ★ 284]

- Extended Ψ_0 's infrastructure into a cross-engine humanoid loco-manipulation benchmark, preparing 1,500+ objects, 50 scenes, and Unitree G1 assets for MuJoCo simulation and Isaac Sim rendering.
- Automated whole-body demonstration generation by integrating BoDEX dexterous grasps with cuRobo motion planning, exporting synchronized visual observations, robot states, and actions in LeRobot format for policy training.
- Enabled VR demonstration collection for complex dexterous tasks beyond automated motion planning, integrating PICO retargeting with GEAR-SONIC and decoupled whole-body control; supported synchronized cross-engine replay of 50-Hz trajectories.
- Built standardized policy evaluation with three progressively challenging domain-randomization levels, varying distractors, lighting, and target-object poses to assess robustness and generalization.

Ψ_0 : An Open Foundation Model Towards Universal Humanoid Loco-Manipulation

Co-author • RSS 2026 • [Project] [Code ★ 2,818]

CVPR 2026 3D-LLM/VLA Workshop Best Paper Award

- Developed a humanoid loco-manipulation benchmark spanning 50+ tasks, 50 scenes, and 70+ objects in Isaac Sim and MuJoCo for training and evaluating robot policies.
- Configured Dex3-1 and Inspire hand assets on Unitree G1 in simulation, including grasp-pose generation and motion-planning pipelines for producing robot trajectories used in policy training.
- Built and executed the policy-evaluation pipeline, benchmarking Ψ_0 against GR00T, InternVLA, ACT, and Diffusion Policy across the full task suite.

Sideways or Away? Socially Aware Robot Navigation via Vision-Based Human-Inspired Imitation Learning

First author • Major revision at ACM Transactions on Human-Robot Interaction (THRI)

- Led the system design and data pipeline, collecting synchronized RGB-D, LiDAR, and odometry streams in ROS and training an ACT-based policy for human-inspired close-range passing.
- Integrated a vision-based decision module with model-based MPPI navigation and conducted real-robot evaluations in narrow-corridor and open-space scenarios.

RESEARCH EXPERIENCE

XVI Robotics

Jul. 2026 – Present

Research Intern; Humanoid Foundation Models and Robot Learning

Beijing, China

- Developing terrain-adaptive whole-body motion tracking with a teacher–student SONIC framework, targeting execution of flat-ground teleoperation references on uneven terrain; implemented and evaluated initial behavior-cloning and student-rollout DAgger pipelines using a terrain-aware teacher.
- Built a simulation pipeline for video-conditioned humanoid inverse-dynamics training, generating synchronized egocentric video and robot state–action data for Unitree G1 with bilateral Inspire RH56 hands; produced 2,300 LeRobot episodes across 23 motion families for humanoid foundation-model pretraining.

Physical Superintelligence (PSI) Lab, University of Southern California

Jul. 2025 – Jun. 2026

Research Intern; Prof. Yue Wang (advisor)

Remote

- Conducted research on Ψ_0 and SIMPLE, contributing to cross-engine simulation benchmarks, scalable demonstration generation, and standardized policy evaluation for humanoid loco-manipulation.

HRI Lab, Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences

Oct. 2024 – Jun. 2025

Research Intern; Prof. Yuyi Liu (advisor)

Shanghai, China

- Studied socially aware navigation on wheeled humanoids using multimodal perception, imitation learning, and model-based motion planning, including adaptive sidestepping in dense and spatially constrained environments.
- Integrated LiDAR, RGB-D, and odometry streams in ROS to build a multimodal data-collection pipeline for real-robot navigation experiments.

PATENT

A Three-Barrel Launching Robot

Aug. 1, 2025

Invention Patent, No. CN120027644B

TECHNICAL SKILLS

Programming & Machine Learning Python, Java, Julia, PyTorch, NumPy

Robotics & Simulation ROS, MuJoCo, NVIDIA Isaac Sim, cuRobo, LeRobot, Unitree SDK

Tools Git, Linux, L^AT_EX